

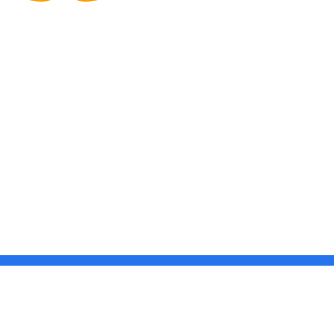
Roadmap to become a NLP Expert in 2025

Prerequisites

- Understanding of Python language
- Understanding of Machine Learning and Deep Learning Concepts



Libraries used



Hugging Face

spaCy

PyTorch



TensorFlow



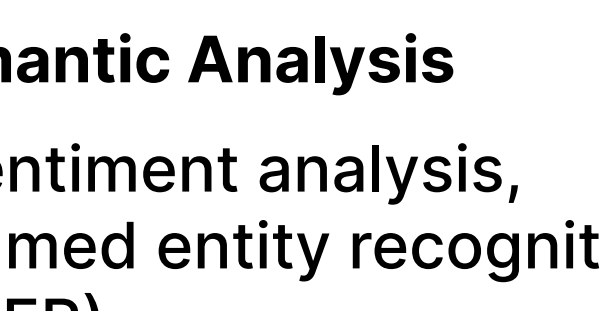
LangChain

Step 1

Core NLP Fundamentals

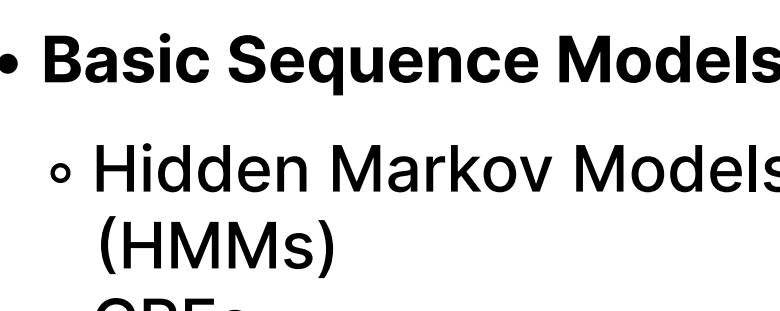
• Text Preprocessing

- Tokenization, stemming, lemmatization.
- Handling stop words, punctuation, and case sensitivity.



• Language Models

- N-grams, Markov models.
- TF-IDF, Bag of Words representation.



• Semantic Analysis

- Sentiment analysis, named entity recognition (NER).
- Part-of-speech tagging.



• Basic Sequence Models

- Hidden Markov Models (HMMs)
- CRFs.



Tools to Learn



spaCy

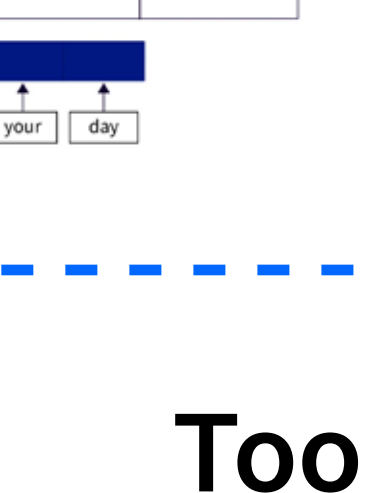


Step 2

Deep Learning for NLP

• Word Embeddings

- Word2Vec, GloVe, FastText.



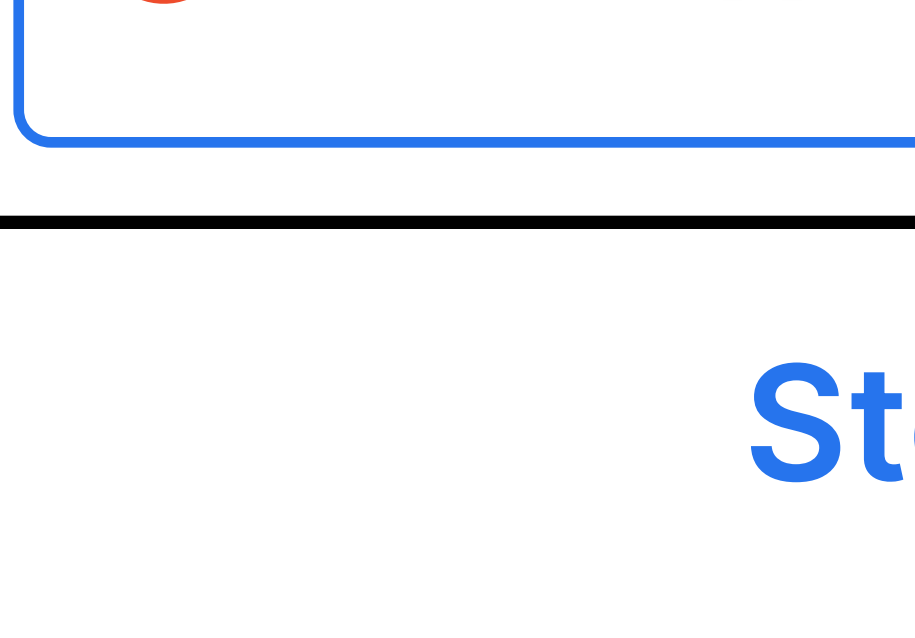
• Recurrent Neural Networks (RNNs)

- GRUs and LSTMs
- Applications in text generation, translation.



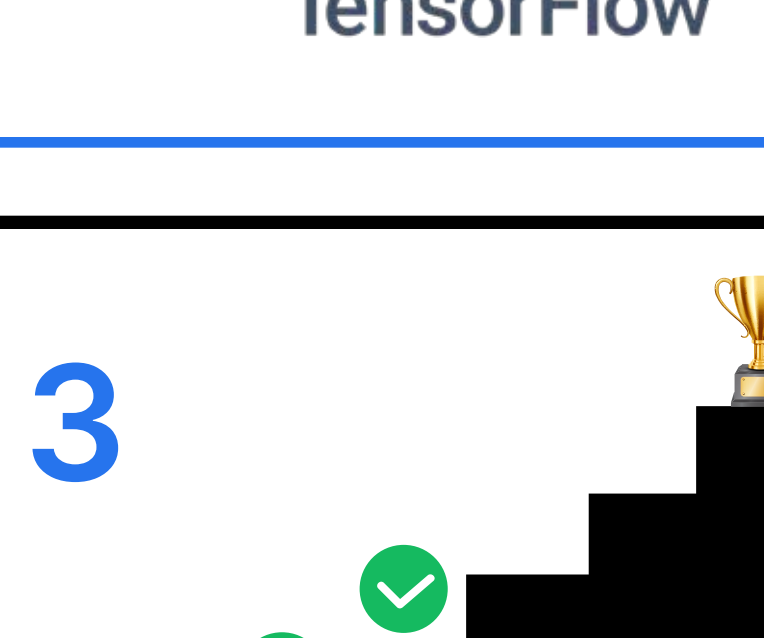
• Attention Mechanisms

- Seq2Seq models with attention.



• Transformers

- Understanding the Transformer architecture.

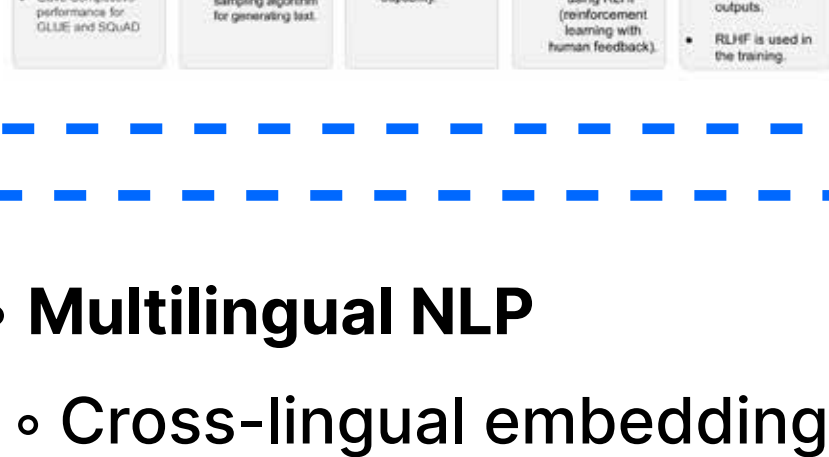


Step 3

Advanced NLP Topics

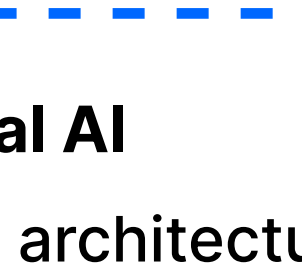
• Pretrained Models

- BERT, GPT series, RoBERTa, T5.



• Text-to-Text Models

- Translation (MarianMT, OpenNMT).
- Summarization (BART, Pegasus).



• Multilingual NLP

- Cross-lingual embeddings.



• Conversational AI

- ChatGPT-like architectures, DialogGPT.
- Fine-tuning conversational models.



Tools to Learn



LangChain



Hugging Face

Step 4

Practical Projects and Applications

• Develop hands-on skills by building

- A chatbot with sentiment detection.
- A document summarization tool.
- An NER system for medical/legal domains.
- A multilingual translation tool.



• Use platforms like Kaggle for NLP competitions.

kaggle

Step 5

Stay Updated and Expand Knowledge

• Follow NLP research and trends

- ArXiv: Weekly read new papers on NLP.
- Blogs: Analytics Vidhya, Hugging Face blog, Distill.pub.
- Communities: Join forums like r/MachineLearning, NLP slack groups.

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